

# Project Report: Development of a Mobile Fitness and Nutrition Tracking Application

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## FitPro



# Introduction

With health problems concerning sedentary behavior, such as obesity and food consumption, being global issues, fitness and nutrition-based tracking application has become a need in society. This project aims to create FitPro a complete mobile solution that enables the users to efficiently log, track, and manage their physical activity, food intake, weight progress, and health progress.

**FitPro** stands out through its unique combination of features :

- Clean, intuitive design that makes fitness and nutrition tracking effortless for users of all levels
- Core functionality available without subscription, ensuring health tracking is available to everyone.
- Evidence-based metrics and recommendations to provide reliable insights into fitness and nutrition.
- Compatibility with major health platforms (Google Fit, Apple Health) for unified data trackin

## 1. Objectives

**primary :** Develop a cross-platform mobile application for fitness/nutrition tracking

**Secondary :**

- Provide actionable statistics
- Integrate wearables and health APIs
- Deliver optimal user experience
- Ensure data confidentiality

## 1. Technical Specifications

### 1. Key Features :

- Secure authentication
- Exercise tracking (type, duration, calories)
- Food journal with nutritional analysis
- Personalized dashboards

### 2. Technical Study :

The technical study represents the phase from design to technical realization. It aims to provide a very detailed functional description of the solution to be implemented, including final process descriptions. The technical study, following the detailed design phase, describes the additional technical specifications needed to complete the realization of the future system. It also describes at a highest level :

- The database information structure
- Program architecture
- Each program's structure and data access methods

For our application development, we utilized several hardware and software resources :

- Programming Language Selection

**Dart - Programming language :**



Dart is specially created for cross-platform applications. This programming language is developed by Google and used for mobile as well as desktop applications and even server and web applications. Dart is an object-oriented language, class-based, garbage-collected with a C-like syntax. It can be compiled to native code or to JavaScript. Dart supports interfaces, abstract classes, reified generics, and type inference.

## **Development Tools Selection :**



Flutter is an SDK for developing graphical user interfaces. It was created by Google and is used to develop applications for Android, iOS, Linux, Mac, Windows, Google Fuchsia, and even webpages using a single codebase. The first version of Flutter was codenamed "Sky" which ran on Android. It was showcased in 2015 at the Dart Developer Summit where it was claimed to perform rendering at 120 FPS. Google announced during the Developer Days Keynote in Shanghai that Flutter Release Preview 2 was issued which was the last major version before Flutter 1.0. Flutter 1.0 was released on 4th December 2018 at the Flutter Live event which marked the first "stable" release of the framework. At the Flutter Interactive event on December 11, 2019, Flutter 1.12 was released.



Android Studio is an Integrated Development Environment (IDE) for creating Android mobile applications. It is built on top of IntelliJ IDEA and employs the Gradle build system. It is available for download on Windows, macOS, Chrome OS, and Linux operating systems. In Android Studio, one can mostly edit Java/Kotlin files, the XML configuration files of an Android application, and the project's build files. It supports the management of multilingual applications and provides the capability to visualize screen layouts on multiple displays of different resolutions at the same time. An emulator that executes a virtual copy of Android on a computer is also part of the package.

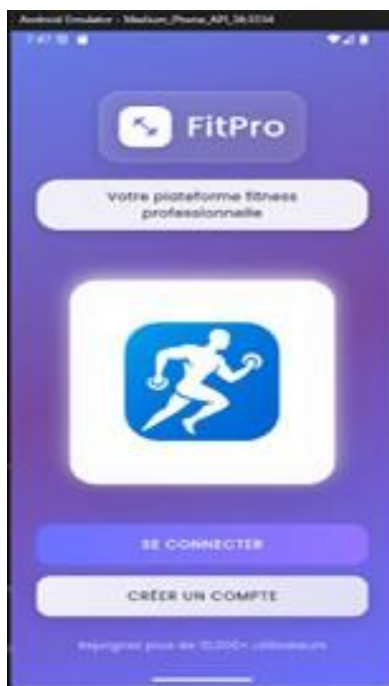
## **3. App Implementation :**

### **1. Development Environment :**

- IDE : Android Studio + VS Code
- Tools :
  - Flutter 3.22
  - Dart 3.2
  - Google Fit API
- Workflow : Git Flow

### 0.3.2 Key Interfaces

#### Home/Authentication Screen



The screenshot indicates two options available on the "Welcome Screen" of the **FitPro** application: "LOG IN" and "CREATE AN ACCOUNT." By selecting "LOG IN," users can enter into their account. A small authentication form will follow from this. To complete the sign in process the user will complete the form with their user profile information ( first name, last name and password). The system's design is simple, task based with an emphasis on familiarity. Users of any experience level will find it easy to navigate the app, due to its minimalistic design.

The image illustrates **FitPro's** login page which includes

sections for username and password, a 'RememberMe'

checkbox, and a large 'SIGN IN' button. There is also a sign up

link available for users who do not have an account. The

aesthetics are simple and uncluttered which emphasizes speed of access while guiding every user through the process

effortlessly with good user flow. The interface aligns with

common mobile login workflows tracking user unlock workflows.





## Main Dashboard Screen :

A dashboard with metrics: today's caloric intake exercise times, workouts completed, and meals tracked. the current date/time with an icon in the top right leading to meal tracker if it is empty. Bottom navigation (Dashboard/Workouts/Meals/Profile) quickly connects users to user-facing key functions.



Additionally, large buttons invite users to track new workouts and meals alike. It is aesthetically pleasing with use of white space, progress bars and typography that cannot get any clearer and easy to read.

## Workout Screen :

The image depicts FitPro's workout tracking screen, showcasing FitPro's simple and modern design style. The key components of the screen are the title "My Workouts", which sets the context of the current page, the week reporting display of statistics, and the options for filtering the workout types (All, Cardio, Strength, HIIT). The center of the screen indicates that the user does not yet have any workouts logged and invites them to Add their first workout with the large "+ Add" button. FitPro provides routine muted bottom navigation (Dashboard, Workouts, Meals, Profile) to provide easy navigation for the rest of the app. The display's simple aesthetic encourages essential functionality with a clear visual hierarchy and simple usability paradigms to support management of workouts.



## Meals Screen :

The meal tracking screen displays a clean screen with the titre "My Meals" along with the timestamp. The meal tracking screen shows a clear empty state, testing "No meals recorded - Start by adding your first meal.' There is a discovery button "Explore Meals (APf)" that suggests meal options for the user. There is a large "+" button at the bottom to aide in recording a new meal needed for tracking This hits on the minimalisñc design focus - directing users' attention to the primary function of meal tracking while creating a step for visually locating where logging food would begin. Overall each aspect of the meal tracking screen fits the minimalist scheme of the app while enabling user interaction expectations.

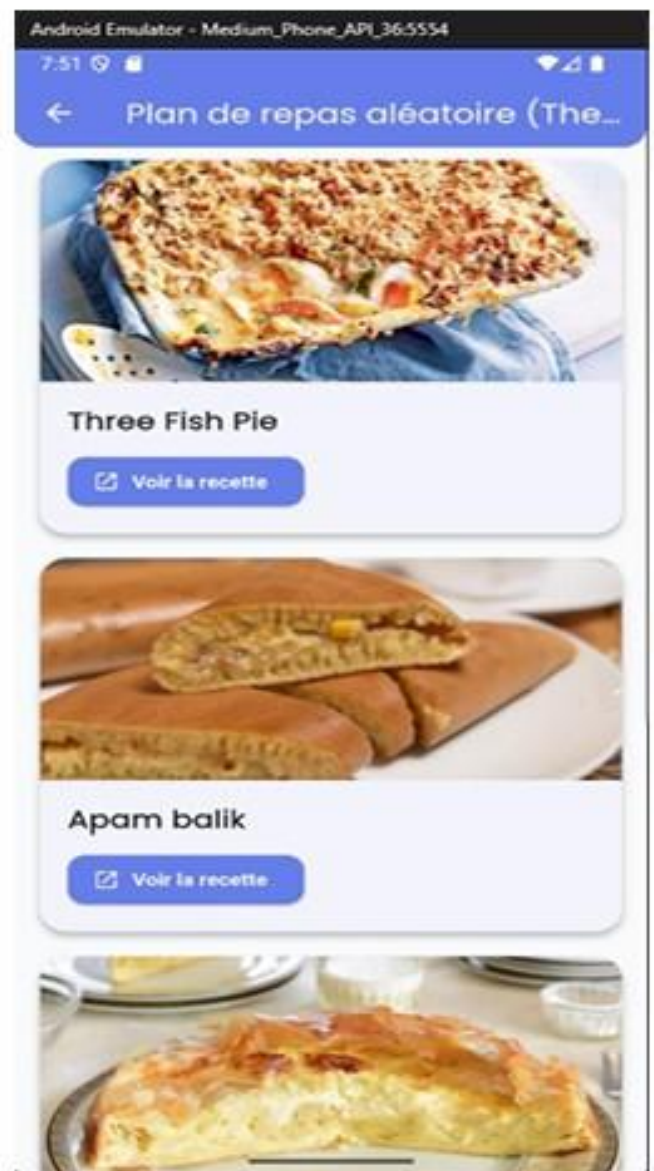


The form is on the screen, which is called "Add a Meal", and it has two options 'New Meal' and "Meal Plan" while collecting fundamental means of nutritional information through clear labels entitled - meal name, calories, macros The form automatically shows what the time is to set the meal timing when they are adding their entry. The form also has a highlighted

"Add Meal' button that indicates they will positively submit their entry. The design and layout is good, made good use of typography, and the fields were intuitively grouped as to focus on constructive nutritional data.



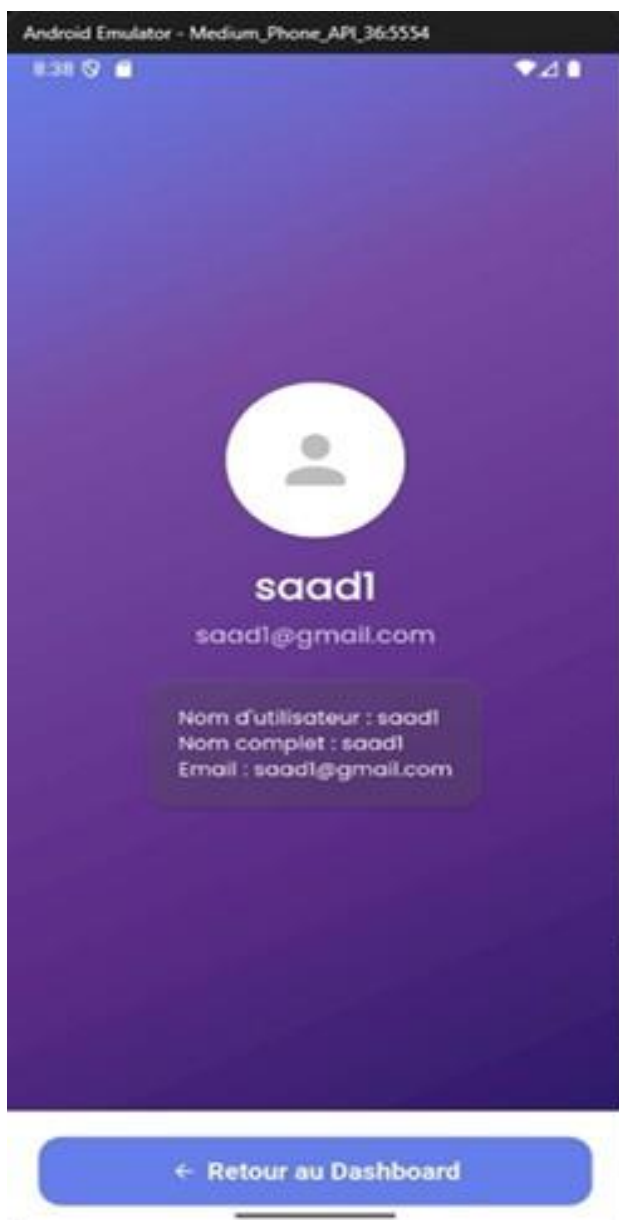
The meal plan screen serves as an interface for the API-based meal recommendation system, and displays the recommendations in a basic card design format. Each recommendation shows the name of the dish and a "View Recipe" option from which the user can go directly to the recipe curated from the integrated nutrition API. The design utilizes a minimal point of view and engages users with nutrition API-derived meal plans. The implementation primarily functions in two ways: first, providing randomized healthy meal ideas, and second, allowing the user to access the details of the recipes immediately. The integration of the API abides by accepted food/nutrition app practices, allowing reliable data delivery while maintaining the simplicity of the app's user experience. The model of the screen is designed to be fast and provide one-touch access into the full details of the recipe provided via the API.



## User Profile Screen :

This screen contains basic account information organized in a structured way, containing the *iaseds* credentials like username, full name, and email address. The interface follows the same design principles/ language as depicted throughout the application that maintains clear typography and hierarchy of information. The tossers are probably on the dashboard and there is a navigation button to get back

there, to keep users within a seamless workflow. It keeps a simple layout, which favour respects the integrity of the information over sheer aesthetics of interface design. We prioritize reading and allowing users to get a quick access to their basic details , while still following the minimalist principles of the application. This screen fulfills the basic premise of allowing users to visualize and validate their account information value, within its application ecosystem.



## 0.4 Conclusion

This project has been not only an academic experience, but it has also been a truly collective experience in the mobile development industry. For the first time, we had the opportunity to work collaboratively on a real product which happened to be a fitness/nutrition tracking mobile application. We had the opportunity to see how a development team works, how to manage version control, how to learn new terms such as Flutter, and how to overcome technical roadblocks. When building the application, we faced roadblocks whether it was learning the tools, understanding dependency deliverables, or integrating poorly documented APIs into our mobile app. All of these roadblocks made us stronger - when it comes to our independence, soft skills like adaptability, analytical skills, but really our teamwork when it comes to working in a team environment in our sector. This project allowed us to display technology skills and further secure the skills we learned in this class, but also allow us to learn new skills as it pertains to cross platform, mobile development capabilities. We were able to learn modern tools and frameworks which we will absolutely take on to future projects.

Lastly, even if the app is working, we want to keep it all going to improve it to be more efficient and more competitive with existing products out there. Completing this project had a unique learning opportunity, and it has been a net positive for our development both professionally and personally.